Ejercitario 1: Conjuntos Difusos y sus Operaciones

Responde a las preguntas de abajo utilizando este mismo notebook. Recuerda de seguir las instrucciones de envío que están en la plataforma Educa.

Ejercicio 1. Definimos un poset $(R^n, \leq)$ donde para todo $x=(x_i)_{i=1}^n, y=(y_i)_{i=1}^n \in R$ tenemos que $x \leq y$ si y solo si $x_i \leq y_i$ para todo $i=1,\ldots,n$. Demuestra que $(R^n, \leq)$ es de hecho un poset.

Respuestas:

Para demostrar que $(R^n, \leq)$ es un poset, se deben cumplir las tres propiedades fundamentales de un poset: reflexividad, antisimetría y transitividad.

Reflexivity: $x \leq x$

Para cualquier elemento $x=(x_i)_{i=1}^n \in R^n$ observamos que cada componente es menor o igual a sí mismo en R, debido a la propiedad reflexiva del orden $\leq$ en R.

Por lo tanto, se cumple la propiedad de reflexividad $x \leq x$.

Transitivity: if $x \leq y$ and $y \leq z$, then $x \leq z$

Teniendo tres elementos $x=(x_i)_{i=1}^n, y=(y_i)_{i=1}^n, z=(z_i)_{i=1}^n \in R^n$ tal que $x \leq y$ y $y \leq z$.

Esto implica que para cada componente i, tenemos $x_i \leq y_i$ y $y_i \leq z_i$ en R.

Dado que el orden $\leq$ en R es transitivo, podemos concluir que $x_i \leq z_i$ para cada i.

Antisymmetry: if $x \leq y$ and $y \leq x$, then $x = y$

Teniendo dos elementos $x=(x_i)_{i=1}^n, y=(y_i)_{i=1}^n \in R^n$ tal que $x \leq y$ y $y \leq x$.

Esto implica que para cada componente i, tenemos $x_i \leq y_i$ y $y_i \leq x_i$ en R.

Dado que el orden $\leq$ en R es antisimétrico, podemos concluir que $x_i = y_i$ para cada i.

Ejercicio 2. Sea $Q[x]$ el conjunto de todos los polinomios de una sola variable x con coeficientes en los números racionales Q. Un polinomio de la forma $a_{n-1}x^{n-1}+\cdots+a_0$ puede representarse como una tupla $(a_{n-1},\ldots,a_0)\in Q^n$.

Considera el conjunto de polinomios de una sola variable con coeficientes racionales no negativos $Q^{+0}[x]$. Decimos que para dos polinomios $p,q\in Q^{+0}[x]$ se tiene que $p\leq q$ si los vectores correspondientes de p y q denotados v_p y v_q se cumple que $v_p\leq v_q$ en el poset $(R,\leq)$ del Ejercicio 1. (i) ¿Cuál es el ínfimo de $(Q^{+0}[x],\leq)$? ¿Porqué?. (ii) ¿Cuál es el supremo de $(Q^{+0}[x],\leq)$? ¿Porqué?. (iii) ¿Tiene $(Q^{+0}[x],\leq)$ elementos mínimos o máximos? Explica porque si o porque no, y en caso positivo escribe explícitamente el mínimo o el máximo.

Ejercicio 3. Demostrar que $([0,1),\leq)$ es un retículo completo y distributivo.

Respuestas:

Un retículo se considera completo si cualquier subconjunto no vacío de él tiene un supremo (límite superior) y un ínfimo (límite inferior) en el retículo.

En el conjunto $[0,1)$, el valor 0 actúa como el ínfimo de cualquier par de elementos y el valor 1 actúa como el supremo de cualquier par de elementos.

Por lo tanto, podemos concluir que $([0,1),\leq)$ **es un retículo completo.**

Para demostrar que el retículo es completamente distributivo, se deben cumplir las leyes de distribución para cualquier elección de elementos A, B y C en el retículo.

Por ejemplo, si $\mu_A=0.2$, $\mu_B=0.3$ y $\mu_C=0.4$ entonces:

(i) $A\vee(B\wedge C)=(A\vee B)\wedge(A\vee C)$

$max(\mu_A,min(\mu_B,\mu_C))=min(max(\mu_A,\mu_B),max(\mu_A,\mu_C))$

$max(0.2,min(0.3,0.4))=min(max(0.2,0.3),max(0.2,0.4))$

$max(0.2,0.3)=min(0.3,0.4)=0.3$

(ii) $A\wedge(B\vee C)=(A\wedge B)\vee(A\wedge C)$

$min(\mu_A,max(\mu_B,\mu_C))=max(min(\mu_A,\mu_B),min(\mu_A,\mu_C))$

$min(0.2,max(0.3,0.4))=max(min(0.2,0.3),min(0.2,0.4))$

$min(0.2,0.4)=max(0.2,0.2)=0.2$

Finalmente podemos concluir que $([0,1),\leq)$ es un retículo completo y distributivo.

Ejercicio 4. Considera los conjuntos difusos $A,B:\{1,2,\ldots,10\}\to[0,1)$ definidos como

$$A(x)=0.2/2+0.7/3+1/4+0.7/5+0.2/6$$

y

$$B(x)=0.3/4+0.5/5+0.8/6+1/7+0.5/8+0.2/9.$$

1. Escribe utilizando la notación de Zadeh (i) $A\cup B$, (ii) $A\cap B$, (iii) $\overline{A}$, (iv) $\overline{B}$, (v) $A\cup\overline{A}$, (vi) $A\cap\overline{A}$, (vii) $\overline{A\cap B}$, (viii) $\overline{A\cup B}$.
2. Dibuja los conjuntos (i) $A\cup B$, (ii) $A\cap B$, (iii) $\overline{A}$, (iv) $\overline{B}$, (v) $A\cup\overline{A}$, (vi) $A\cap\overline{A}$, (vii) $\overline{A\cap B}$, (viii) $\overline{A\cup B}$.
3. Escribe formalmente el soporte (*support*), núcleo (*core*) y altura (*height*) de A y B.

Respuestas:

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1. Escribe utilizando la notación de Zadeh

(i) $A\cup B(x)=0.2/2+0.7/3+1/4+0.7/5+0.8/6+1/7+0.5/8+0.2/9$

(ii) $A\cap B(x)=0.3/4+0.5/5+0.2/6$

(iii) $\overline{A}(x)=1/1+0.8/2+0.3/3+0.3/5+0.8/6+1/7+1/8+1/9+1/10$

(iv) $\overline{B}(x)=1/1+1/2+1/3+0.7/4+0.5/5+0.2/6+0.5/8+0.8/9+1/10$

(v) $A\cup\overline{A}(x)=1/1+0.8/2+0.7/3+1/4+0.7/5+0.8/6+1/7+1/8+1/9+1/10$

(vi) $A\cap\overline{A}(x)=0.2/2+0.3/3+0.3/5+0.2/6$

(vii) $\overline{A\cap B}(x)=1/1+1/2+1/3+0.7/4+0.5/5+0.8/6+1/7+1/8+1/9+1/10$

(viii) $\overline{A\cup B}(x)=1/1+0.8/2+0.3/3+0.3/5+0.2/6+0.5/8+0.8/9+1/10$

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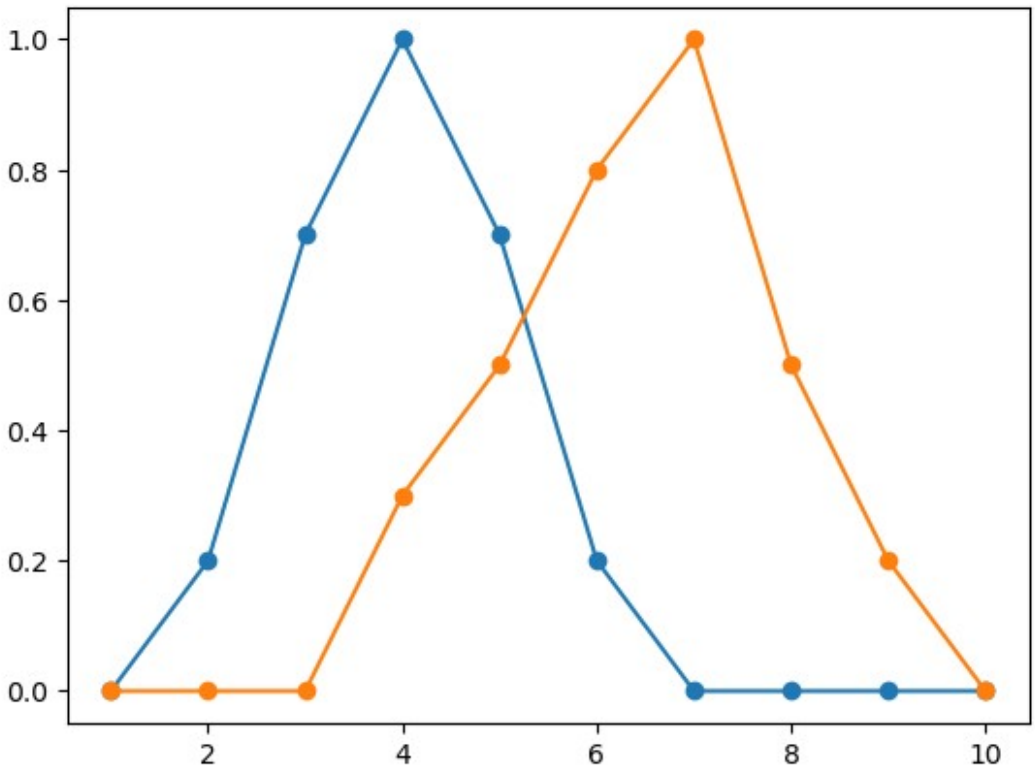
2. Dibuja los conjuntos

Generé un gráfico para visualizar los conjuntos A y B:

```
import matplotlib.pyplot as plt

x=[1,2,3,4,5,6,7,8,9,10]
a=[0,0.2,0.7,1,0.7,0.2,0,0,0,0]
b=[0,0,0,0.3,0.5,0.8,1,0.5,0.2,0]
```

```
plt.plot(x,a, "-o")
plt.plot(x,b, "-o")
plt.show()
```

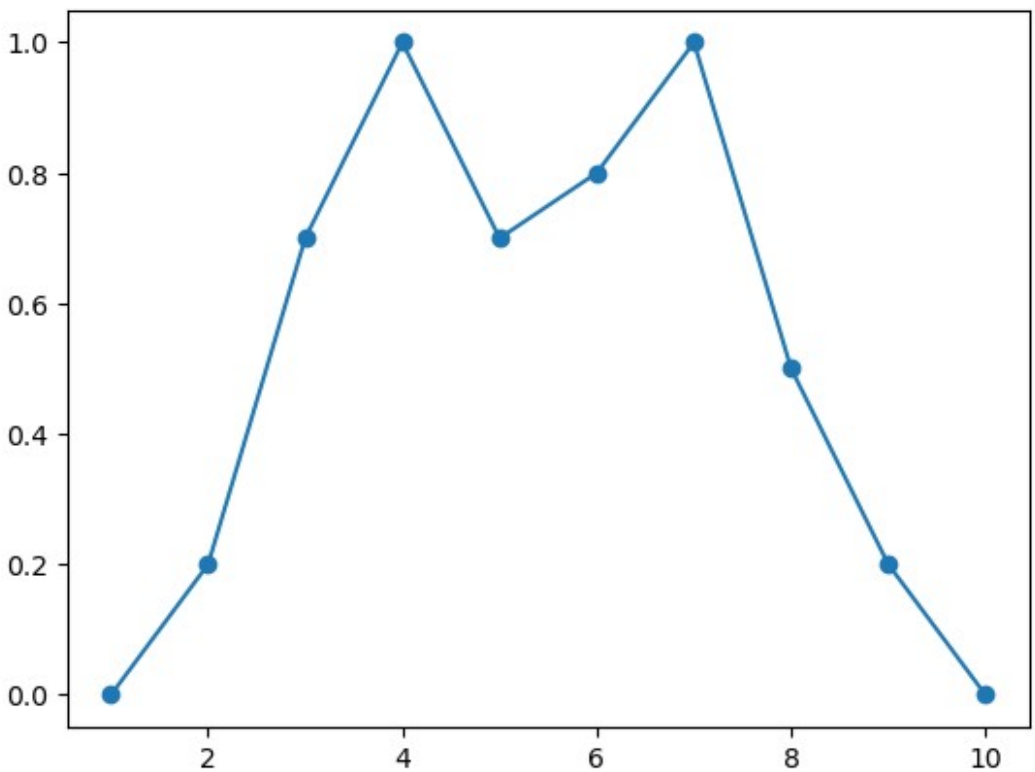


(i) $A \cup B$

```
import matplotlib.pyplot as plt

x=[1,2,3,4,5,6,7,8,9,10]
y=[0,0.2,0.7,1,0.7,0.8,1,0.5,0.2,0]

plt.plot(x,y, "-o")
plt.show()
```

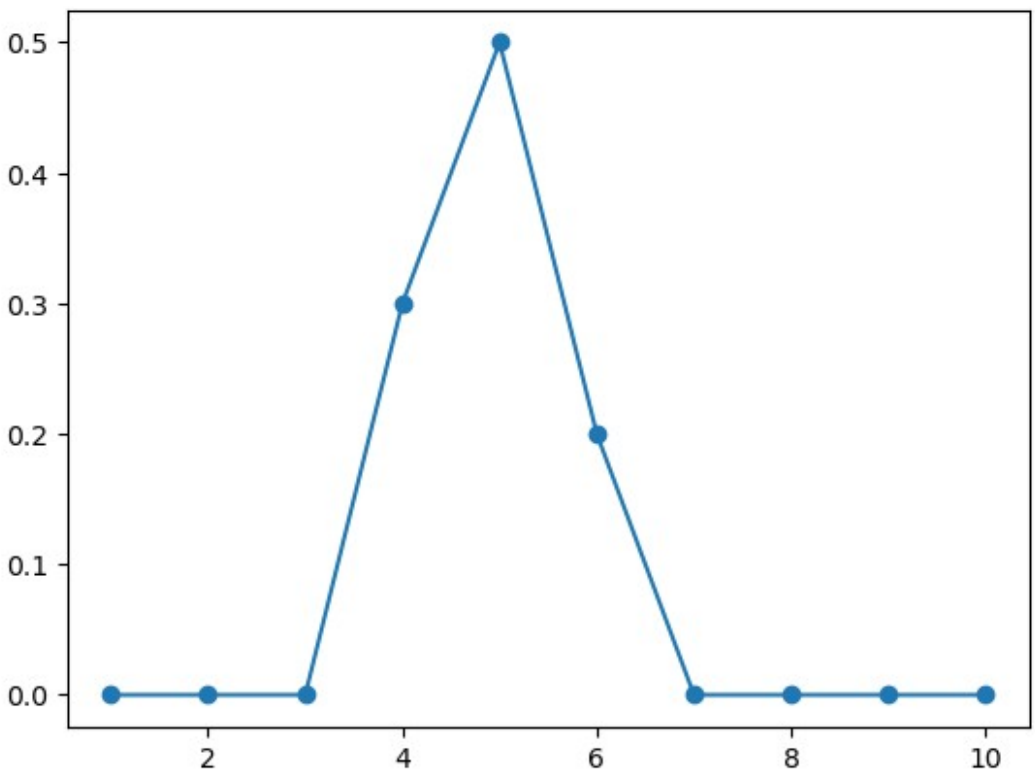


(ii) $A \cap B$

```
import matplotlib.pyplot as plt

x=[1,2,3,4,5,6,7,8,9,10]
y=[0,0,0,0.3,0.5,0.2,0,0,0,0]

plt.plot(x,y, "-o")
plt.show()
```

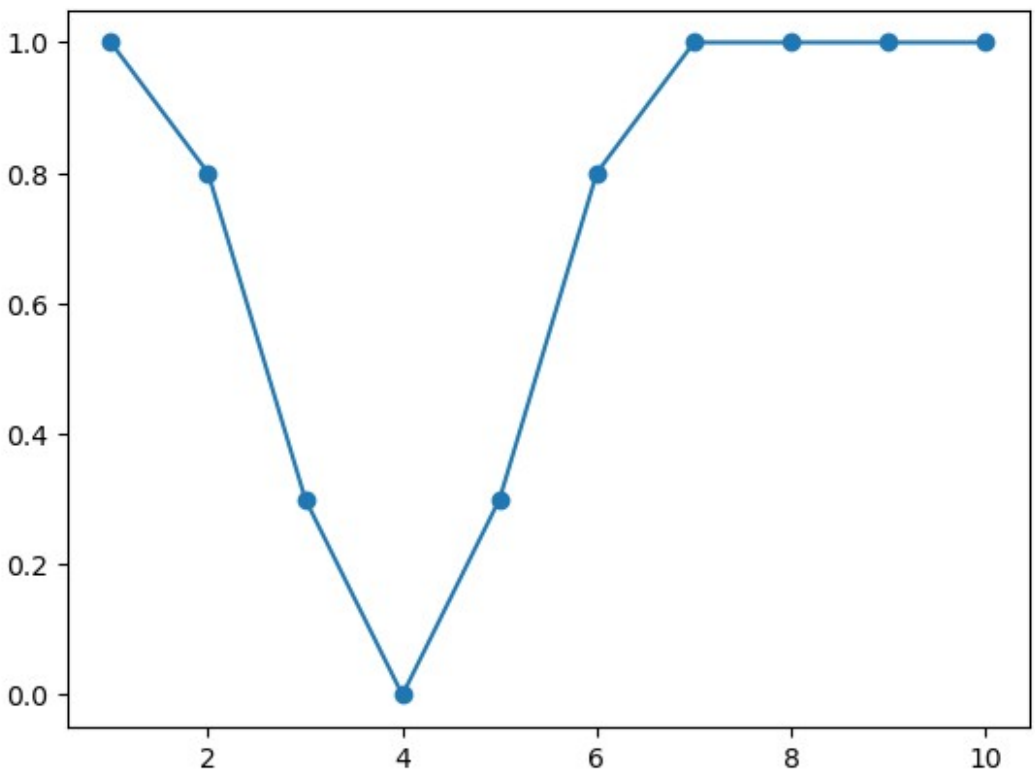


(iii) $\overline{A}$

```
import matplotlib.pyplot as plt

x=[1,2,3,4,5,6,7,8,9,10]
y=[1,0.8,0.3,0,0.3,0.8,1,1,1,1]

plt.plot(x,y, "-o")
plt.show()
```

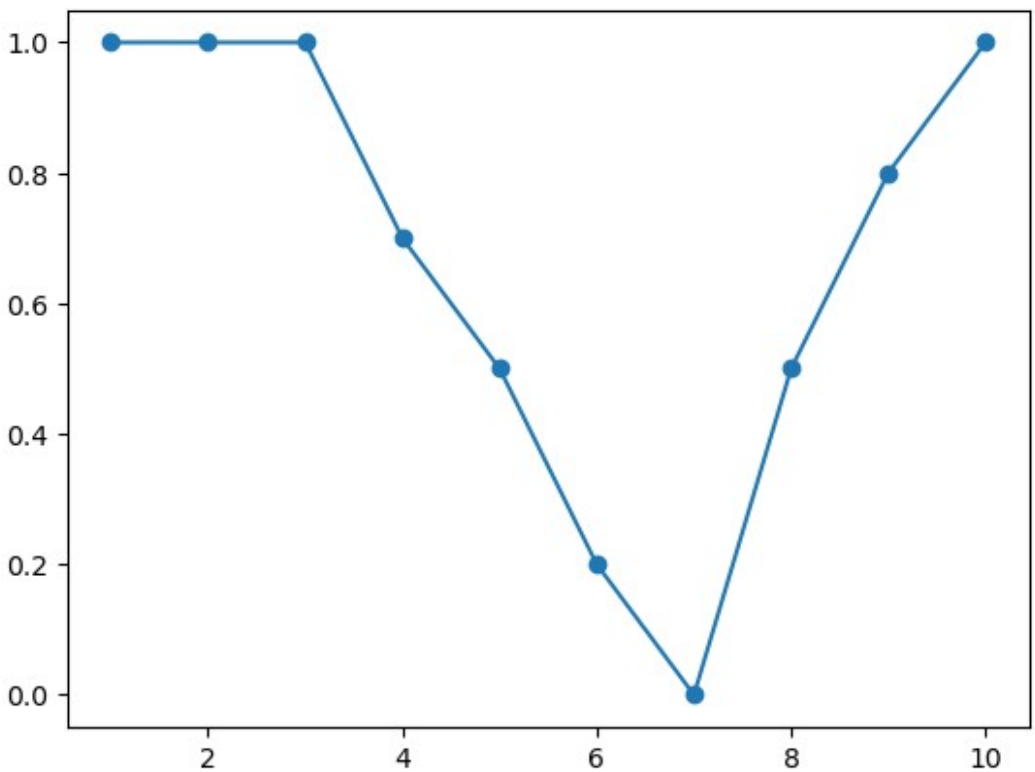


(iv) $\overline{B}$

```
import matplotlib.pyplot as plt

x=[1,2,3,4,5,6,7,8,9,10]
y=[1,1,1,0.7,0.5,0.2,0,0.5,0.8,1]

plt.plot(x,y, "-o")
plt.show()
```

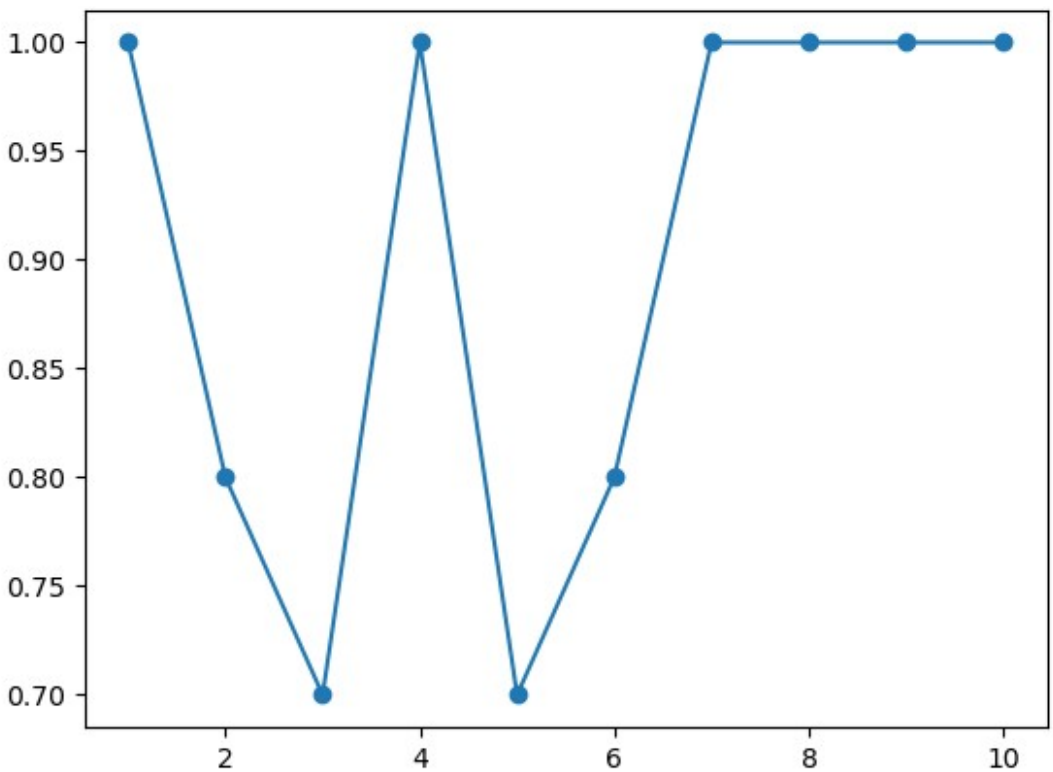


(v) $A \cup \overline{A}$

```
import matplotlib.pyplot as plt

x=[1,2,3,4,5,6,7,8,9,10]
y=[1,0.8,0.7,1,0.7,0.8,1,1,1,1]

plt.plot(x,y, "-o")
plt.show()
```

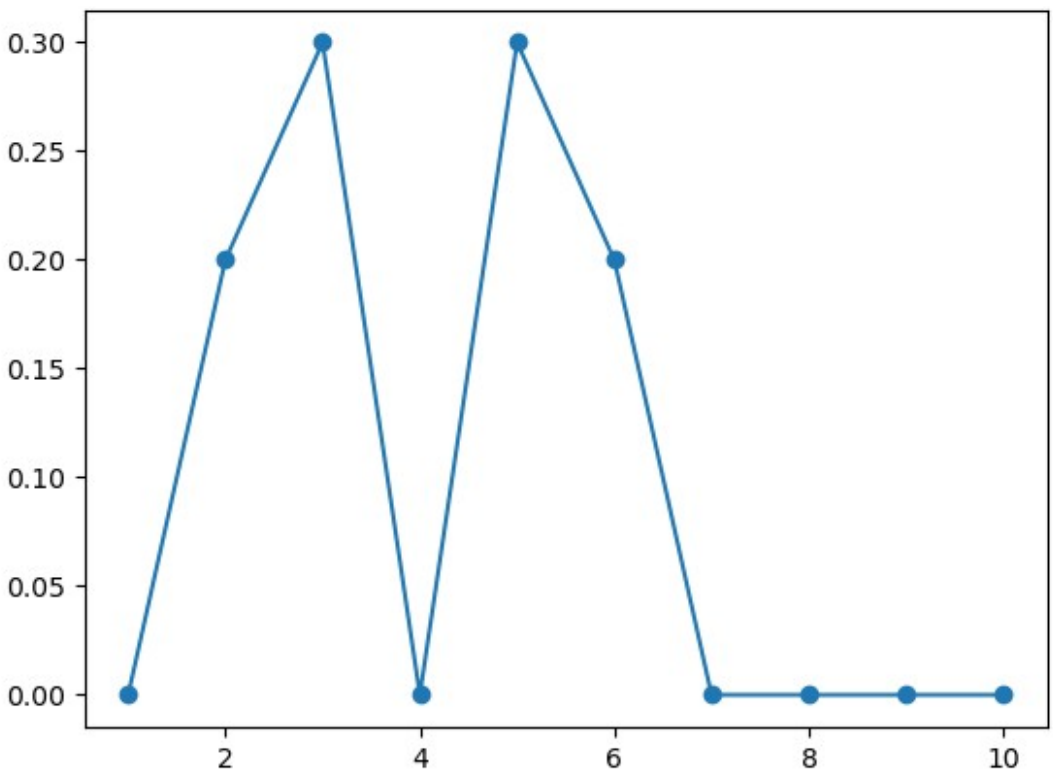


(vi) $A \cap \overline{A}$

```
import matplotlib.pyplot as plt

x=[1,2,3,4,5,6,7,8,9,10]
y=[0,0.2,0.3,0,0.3,0.2,0,0,0,0]

plt.plot(x,y, "-o")
plt.show()
```

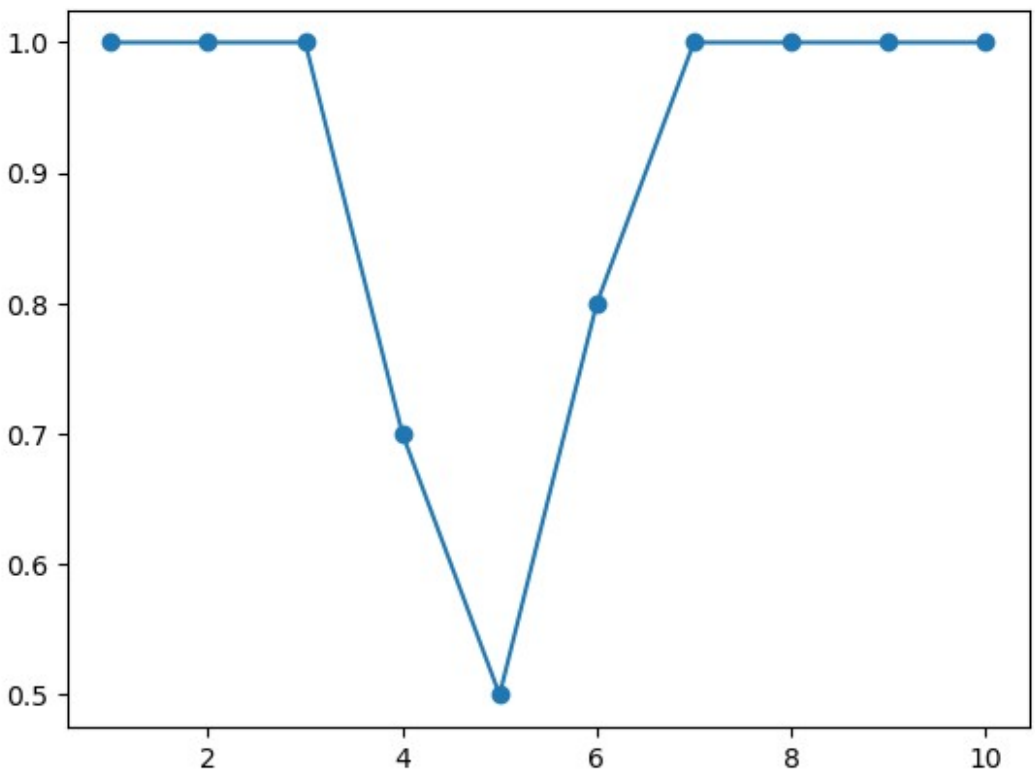


(vii) $\overline{A \cap B}$

```
import matplotlib.pyplot as plt

x=[1,2,3,4,5,6,7,8,9,10]
y=[1,1,1,0.7,0.5,0.8,1,1,1,1]

plt.plot(x,y, "-o")
plt.show()
```

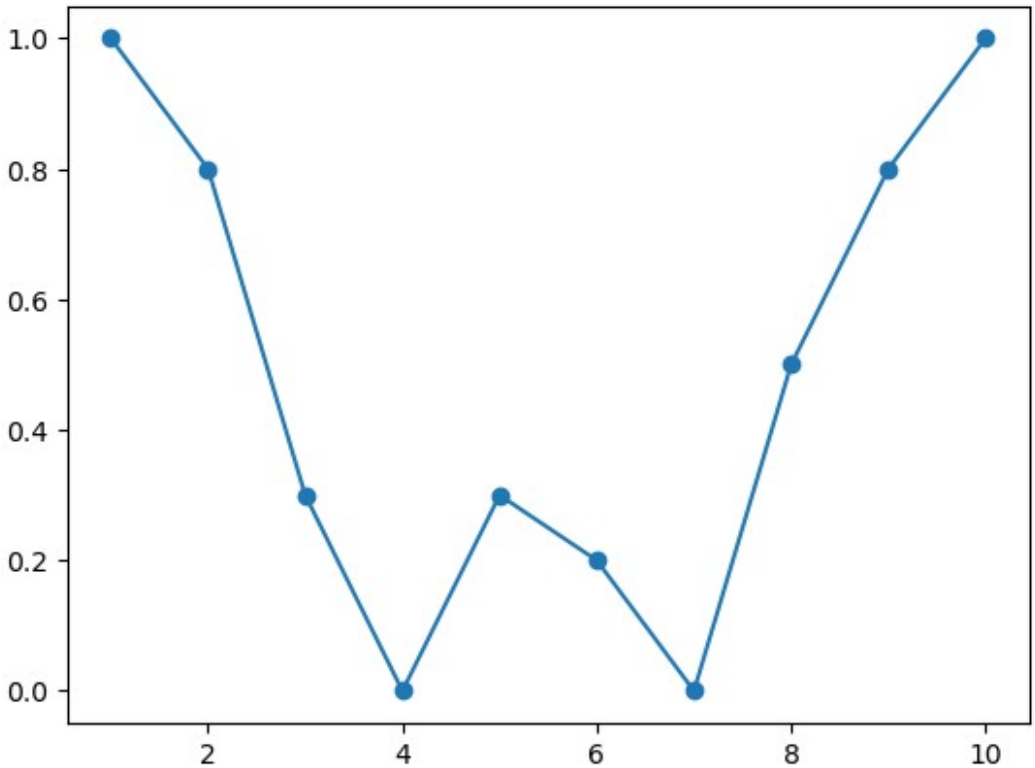


(viii) $\overline{A \cup B}$

```
import matplotlib.pyplot as plt

x=[1,2,3,4,5,6,7,8,9,10]
y=[1,0.8,0.3,0,0.3,0.2,0,0.5,0.8,1]

plt.plot(x,y, "-o")
plt.show()
```



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3. Escribe formalmente el

(i) Soporte (*support*) de A y B

$supp(A)=\{2,3,4,5,6\}$

$supp(B)=\{4,5,6,7,8,9\}$

(ii) Núcleo (*core*) de A y B

$core(A)=\{4\}$

$core(B)=\{7\}$

(iii) Altura (*height*) de A y B

$h(A)=1$

$h(B)=1$

Ejercicio 5. Considera los conjuntos difusos $A, B: R^{+¿} \to [0,1]¿$ definidos como $A(x)=\frac{1}{1+x^2}$ y $B(x)=\frac{1}{10^x}$.

1. Escribe utilizando la notación de Zadeh (i) $A \cup B$, (ii) $A \cap B$, (iii) $\overline{A}$, (iv) $\overline{B}$, (v) $A \cup \overline{A}$, (vi) $A \cap \overline{A}$, (vii) $\overline{A \cap B}$, (viii) $\overline{A \cup B}$
2. Escribe las funciones de membresía de (i) $A \cup B$, (ii) $A \cap B$, (iii) $\overline{A}$, (iv) $\overline{B}$, (v) $A \cup \overline{A}$, (vi) $A \cap \overline{A}$, (vii) $\overline{A \cap B}$, (viii) $\overline{A \cup B}$.
3. Escribe formalmente el soporte (*support*), núcleo (*core*) y altura (*height*) de A y B.

Respuestas:

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1. Escribe utilizando la notación de Zadeh

(i) $A \cup B(x)=max\{\frac{1}{1+x^2},\frac{1}{10^x}\}=\{\int_{R^{+¿}\frac{\frac{1}{1+x^2}}{x}\}¿}$

(ii) $A \cap B(x)=min\{\frac{1}{1+x^2},\frac{1}{10^x}\}=\{\int_{R^{+¿}\frac{\frac{1}{10^x}}{x}\}¿}$

(iii) $\overline{A}(x)=1-\frac{1}{1+x^2}=\{\int_{R^{+¿}\frac{1-\frac{1}{1+x^2}}{x}\}¿}$

(iv) $\overline{B}(x)=1-\frac{1}{10^x}=\{\int_{R^{+¿}\frac{1-\frac{1}{10^x}}{x}\}¿}$

(v) $A \cup \overline{A}(x)=max\{\frac{1}{1+x^2},1-\frac{1}{1+x^2}\}=\{\int_{R^{+¿}\frac{1-\frac{1}{1+x^2}}{x}\}¿}$

(vi) $A \cap \overline{A}(x)=min\{\frac{1}{1+x^2},1-\frac{1}{1+x^2}\}=\{\int_{R^{+¿}\frac{\frac{1}{1+x^2}}{x}\}¿}$

(vii) $\overline{A \cap B}(x)=1-min\{\frac{1}{1+x^2},\frac{1}{10^x}\}=1-\frac{1}{10^x}=\{\int_{R^{+¿}\frac{1-\frac{1}{10^x}}{x}\}¿}$

(viii) $\overline{A \cup B}(x)=1-max\{\frac{1}{1+x^2},\frac{1}{10^x}\}=1-\frac{1}{1+x^2}=\{\int_{R^{+¿}}^{\frac{1-\frac{1}{1+x^2}}{x}\}¿}¿$

---

2. Escribe las funciones de membresía de

(i) $A \cup B(x)=\frac{1}{1+x^2} \vee \frac{1}{10^x}=\frac{1}{1+x^2}$

(ii) $A \cap B(x)=\frac{1}{1+x^2} \wedge \frac{1}{10^x}=\frac{1}{10^x}$

(iii) $\overline{A}(x)=1-\frac{1}{1+x^2}$

(iv) $\overline{B}(x)=1-\frac{1}{10^x}$

(v) $A \cup \overline{A}(x)=\frac{1}{1+x^2} \vee \left(1-\frac{1}{1+x^2}\right)=1-\frac{1}{1+x^2}$

(vi) $A \cap \overline{A}(x)=\frac{1}{1+x^2} \wedge \left(1-\frac{1}{1+x^2}\right)=\frac{1}{1+x^2}$

(vii) $\overline{A \cap B}(x)=1-¿ \wedge \frac{1}{10^x}¿=1-\frac{1}{10^x}$

(viii) $\overline{A \cup B}(x)=1-¿ \vee \frac{1}{10^x}¿=1-\frac{1}{1+x^2}$

---

3. Escribe formalmente el

(i) Soporte (*support*) de A y B

$supp(A)=R^{+¿¿}$

$supp(B)=R^{+¿¿}$

(ii) Núcleo (*core*) de A y B

$core(A)=\varnothing$

$core(B)=\varnothing$

(iii) Altura (*height*) de A y B

$h(A)=1$

$h(B)=1$

Ejercicio 6. Considera los conjuntos difusos A, B definidos como

$$A(x)=\left(\begin{array}{ll} 0 & \text{si } x<1 \\ \frac{x-1}{6} & \text{si } 1\leq x<7 \\ \frac{10-x}{3} & \text{si } 7\leq x<10 \\ 0 & \text{si } 10\leq x, \end{array}\right)$$

y

$$B(x)=\left(\begin{array}{ll} 0 & \text{si } x<2 \\ x-2 & \text{si } 2\leq x<3 \\ 1 & \text{si } 3\leq x<4 \\ \frac{6-x}{2} & \text{si } 4\leq x<6 \\ 0 & \text{si } 6<x, \end{array}\right)$$

1-). Escribe las funciones de membresía de (i) $A \cup B$, (ii) $A \cap B$, (iii) $\overline{A}$, (iv) $\overline{B}$, (v) $A \cup \overline{A}$, (vi) $A \cap \overline{A}$, (vii) $\overline{A \cap B}$, (viii) $\overline{A \cup B}$.

Hay que considerar los intervalos donde cada función es definida y combinarlos para formar una única función.

1) $0=¿x<1$

2) Intersección entre $\frac{x-1}{6}=0$

$ x - 1 = 0$
$ x = 1$

2) Intersección entre $\frac{x-1}{6}=x-2$

$\frac{x-1}{6}=x-2$
$ x-1 = (x-2)6$
$ x-1 = 6x - 12$
$ 11 = 5x$
$ x = \frac{11}{5} = 2.2$

3) Intersección entre $\frac{x-1}{6}=1$

$\frac{x-1}{6}=1$
$ x-1 = 6$
$ x = 7 $

4) Intersección entre $\frac{x-1}{6}=\frac{6-x}{2}$

$\frac{x-1}{6}=\frac{6-x}{2}$
$ 2(x-1) = 6(6-x)$
$ 2x - 2 = 36 - 6x $
$ 8x = 38 $
$ x = 4.75 $

5) Intersección entre $\frac{x-1}{6}=1$

$\frac{x-1}{6}=1$
$ x-1 = 6$
$ x = 7 $

Entonces

(i) $A \cup B$

$$A \cup B(x)=\left(\begin{array}{ll} 0 & \text{si } x<1 \\ \frac{x-1}{6} & \text{si } 1 \leq x<2.2 \\ x-2 & \text{si } 2.2 \leq x<3 \\ 1 & \text{si } 3 \leq x<4 \\ \frac{6-x}{2} & \text{si } 4 \leq x<4.75 \\ \frac{x-1}{6} & \text{si } 4.75 \leq x<7 \\ \frac{10-x}{3} & \text{si } 7 \leq x<10 \\ 0 & \text{si } x \geq 10, \end{array}\right)$$

(ii) $A \cap B$

$$A \cap B(x) = \begin{cases} 0 & \text{si } x < 2 \\ x - 2 & \text{si } 2 \leq x < 2.2 \\ \frac{x-1}{6} & \text{si } 2.2 \leq x < 4.75 \\ \frac{6-x}{2} & \text{si } 4.75 \leq x < 6 \\ 0 & \text{si } x \geq 6, \end{cases}$$

(iii) $\overline{A}$

$$\overline{A}(x) = \begin{cases} 1 & \text{si } x < 1 \\ 1 - \frac{x-1}{6} & \text{si } 1 \leq x < 7 \\ 1 - \frac{10-x}{3} & \text{si } 7 \leq x < 10 \\ 1 & \text{si } 10 \leq x, \end{cases}$$

(iv) $\overline{B}$

$$\overline{B}(x) = \begin{cases} 1 & \text{si } x < 2 \\ 1 - (x-2) & \text{si } 2 \leq x < 3 \\ 0 & \text{si } 3 \leq x < 4 \\ 1 - \frac{6-x}{2} & \text{si } 4 \leq x < 6 \\ 1 & \text{si } 6 < x, \end{cases}$$

(v) $A \cup \overline{A}$

- Intersección entre $1 - \frac{x-1}{6} = \frac{x-1}{6}$

$1 - = 2\left(\frac{x-1}{6}\right)$
$6 = 2x - 2$
$8 = 2x$
$4 = x$

- Intersección entre $1 - \frac{10-x}{3} = \frac{10-x}{3}$

$1 - = 2\left(\frac{10-x}{3}\right)$
$3 = 20 - 2x$
$2x = 17$
$x = 9$

Entonces

$$A \cup \overline{A} = \begin{cases} 1 & \text{si } x < 1 \\ 1 - \dfrac{x-1}{6} & \text{si } 1 \le x < 4 \\ \dfrac{x-1}{6} & \text{si } 4 \le x < 7 \\ \dfrac{10-x}{3} & \text{si } 7 \le x < 8.5 \\ 1 - \dfrac{10-x}{3} & \text{si } 8.5 \le x < 10 \\ 1 & \text{si } 10 \le x, \end{cases}$$

(vi) $A \cap \overline{A}$

$$A \cup \overline{A} = \begin{cases} 0 & \text{si } x < 1 \\ \dfrac{x-1}{6} & \text{si } 1 \le x < 4 \\ 1 - \dfrac{x-1}{6} & \text{si } 4 \le x < 7 \\ 1 - \dfrac{10-x}{3} & \text{si } 7 \le x < 8.5 \\ \dfrac{10-x}{3} & \text{si } 8.5 \le x < 10 \\ 0 & \text{si } 10 \le x, \end{cases}$$

(vii) $\overline{A \cup B}$

$$\overline{A \cup B} = \begin{cases} 1 & \text{si } x < 1 \\ 1 - \dfrac{x-1}{6} & \text{si } 1 \le x < 2.2 \\ 1 - (x-2) & \text{si } 2.2 \le x < 3 \\ 0 & \text{si } 3 \le x < 4 \\ 1 - \dfrac{6-x}{2} & \text{si } 4 \le x < 4.75 \\ 1 - \dfrac{x-1}{6} & \text{si } 4.75 \le x < 7 \\ 1 - \dfrac{10-x}{3} & \text{si } 7 \le x < 10 \\ 1 & \text{si } x \ge 10, \end{cases}$$

(viii) $\overline{A \cap B}$

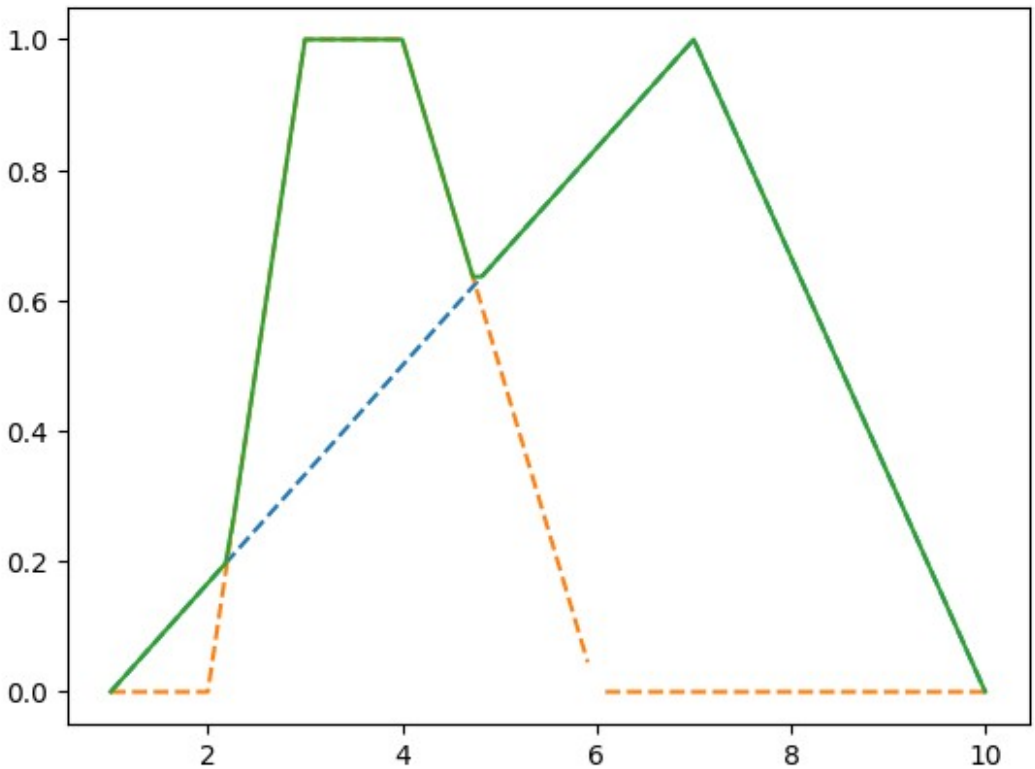
$$\overline{A \cap B}(x) = \begin{cases} 1 & \text{si } x < 2 \\ 1-(x-2) & \text{si } 2 \le x < 2.2 \\ 1-\dfrac{x-1}{6} & \text{si } 2.2 \le x < 4.75 \\ 1-\dfrac{6-x}{2} & \text{si } 4.75 \le x < 6 \\ 1 & \text{si } x \ge 6, \end{cases}$$

1. Dibuja los conjuntos (i) $A \cup B$, (ii) $A \cap B$, (iii) $\overline{A}$, (iv) $\overline{B}$, (v) $A \cup \overline{A}$, (vi) $A \cap \overline{A}$, (vii) $\overline{A \cap B}$, (viii) $\overline{A \cup B}$.
2. Escribe formalmente el soporte (*support*), núcleo (*core*) y altura (*height*) de A y B.

```
from matplotlib import pyplot
import numpy as np

def fA(x):
    if(x<1):
        return 0
    elif  1 <= x and x < 7:
        return (x - 1)/6
    elif  1 <= x and x < 10:
        return (10 - x)/3
    elif x >= 10:
        return 0

def fB(x):
    if(x<2):
        return 0
    elif  1 <= x and x < 2:
        return (x - 1)/6
    elif  2 <= x and x < 3:
        return (x-2)
    elif  3 <= x and x < 4:
        return 1
    elif  4 <= x and x < 6:
        return (6-x)/2
    elif x >6:
        return 0

def fC(x):
    if x < 1:
        return 0
    elif  1 <= x and x < 2.2:
        return (x - 1)/6
    elif  2.2 <= x and x < 3:
        return (x-2)
    elif  3 <= x and x < 4:
        return 1
    elif  4 <= x and x < 4.75:
```

```
        return (6-x)/2
    elif  4.75 <= x and x < 7:
        return (x - 1)/6
    elif  7 <= x and x < 10:
        return (10 - x)/3
    elif x >= 10:
        return 0

x = np.linspace(1, 10, 100)
pyplot.plot(x, [fA(i) for i in x],'--')
pyplot.plot(x, [fB(i) for i in x],'--')
pyplot.plot(x, [fC(i) for i in x],'-')
```

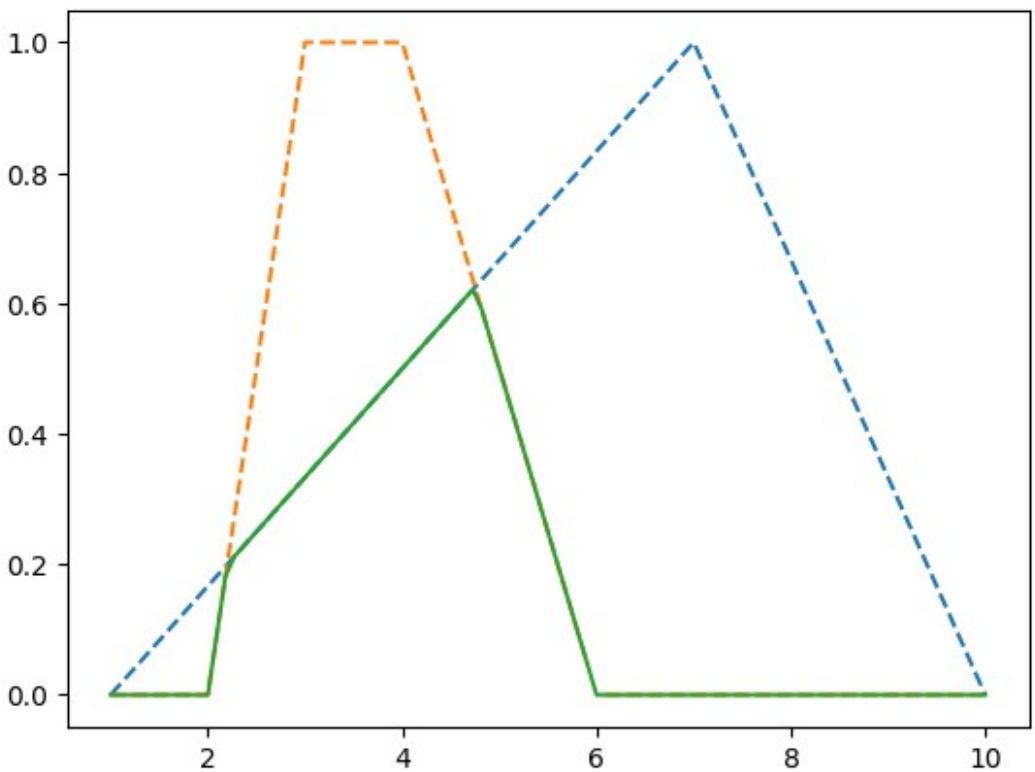
```
[<matplotlib.lines.Line2D at 0x1ad170e9150>]
```




```
from matplotlib import pyplot
import numpy as np

def fA(x):
    if(x<1):
        return 0
    elif  1 <= x and x < 7:
        return (x - 1)/6
    elif  1 <= x and x < 10:
        return (10 - x)/3
```

```
    elif x >= 10:
        return 0

def fB(x):
    if(x<2):
        return 0
    elif  1 <= x and x < 2:
        return (x - 1)/6
    elif  2 <= x and x < 3:
        return (x-2)
    elif  3 <= x and x < 4:
        return 1
    elif  4 <= x and x < 6:
        return (6-x)/2
    elif x >6:
        return 0

def fC(x):
    if x < 2:
        return 0
    elif  2 <= x and x < 2.2:
        return (x -2)
    elif  2.2 <= x and x < 4.75:
        return (x - 1)/6
    elif  4.75 <= x and x < 6:
        return (6-x)/2
    elif x >= 6:
        return 0

x = np.linspace(1, 10, 100)
pyplot.plot(x, [fA(i) for i in x],'--')
pyplot.plot(x, [fB(i) for i in x],'--')
pyplot.plot(x, [fC(i) for i in x],'-')
```

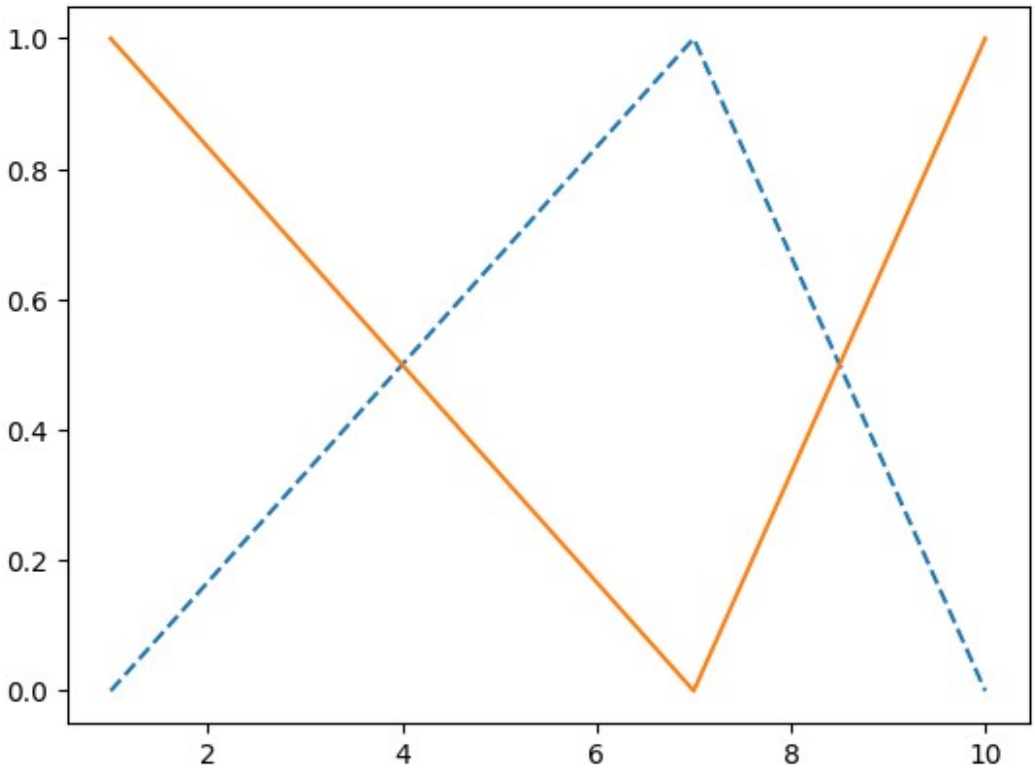
```
[<matplotlib.lines.Line2D at 0x1ad17454e90>]
```




```
from matplotlib import pyplot
import numpy as np

def fA(x):
    if(x<1):
        return 0
    elif  1 <= x and x < 7:
        return (x - 1)/6
    elif  7 <= x and x < 10:
        return (10 - x)/3
    elif x >= 10:
        return 0

def fAA(x):
    if(x<1):
        return 1
    elif  1 <= x and x < 7:
        return 1 - ((x - 1)/6)
    elif  7 <= x and x < 10:
        return 1 - ((10 - x)/3)
    elif x >= 10:
        return 1

x = np.linspace(1, 10, 100)
```

```
pyplot.plot(x, [fA(i) for i in x],'--')
pyplot.plot(x, [fAA(i) for i in x],'-')
```

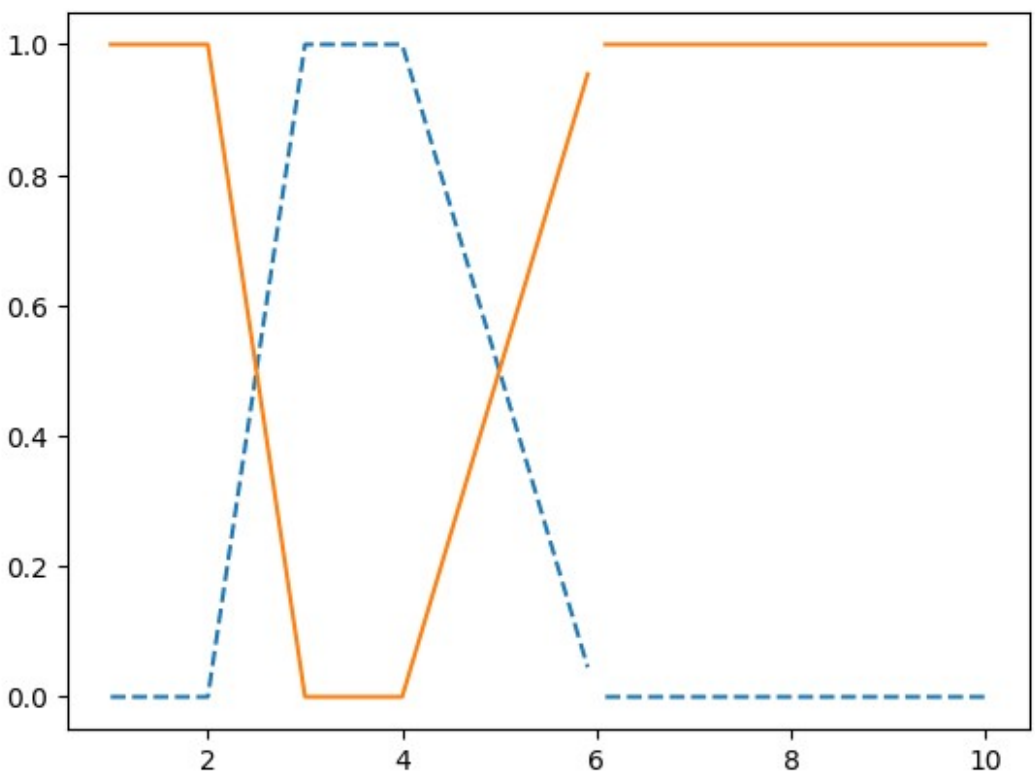
```
[<matplotlib.lines.Line2D at 0x1ad17493d10>]
```




```
from matplotlib import pyplot
import numpy as np

def fB(x):
    if(x<2):
        return 0
    elif  1 <= x and x < 2:
        return (x - 1)/6
    elif  2 <= x and x < 3:
        return (x-2)
    elif  3 <= x and x < 4:
        return 1
    elif  4 <= x and x < 6:
        return (6-x)/2
    elif x >6:
        return 0

def fBB(x):
    if(x<2):
        return 1
```

```
    elif  1 <= x and x < 2:
        return 1-(x - 1)/6
    elif  2 <= x and x < 3:
        return 1-(x-2)
    elif  3 <= x and x < 4:
        return 0
    elif  4 <= x and x < 6:
        return 1-(6-x)/2
    elif x >6:
        return 1


x = np.linspace(1, 10, 100)
pyplot.plot(x, [fB(i) for i in x],'--')
pyplot.plot(x, [fBB(i) for i in x],'-')
```

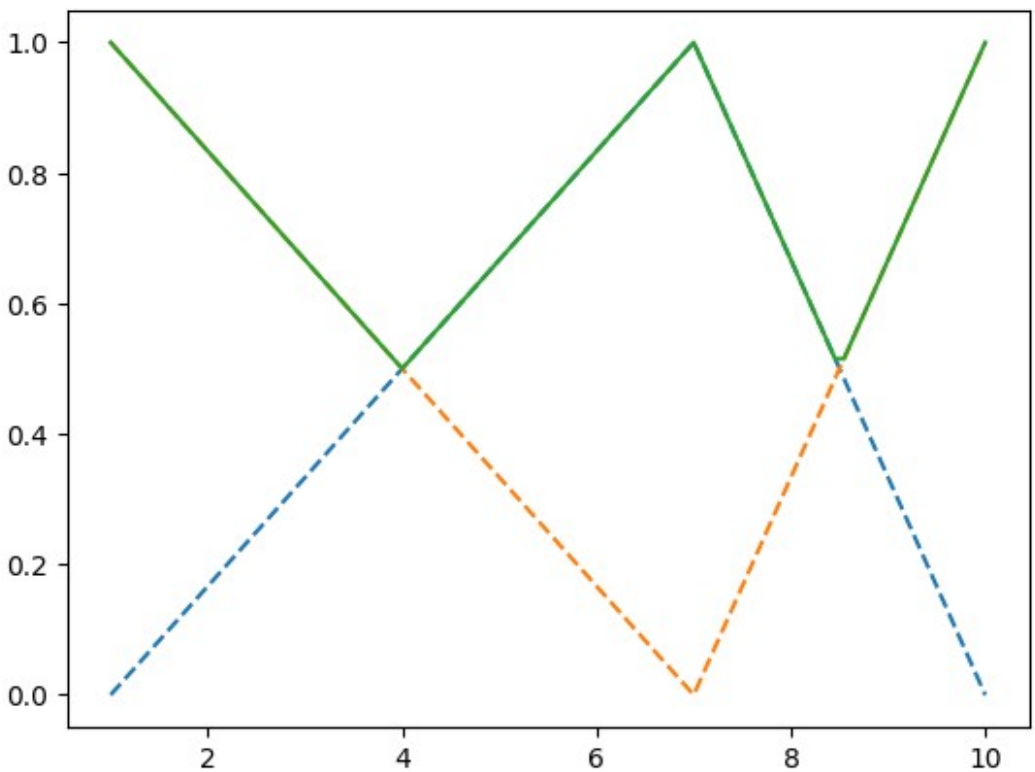
```
[<matplotlib.lines.Line2D at 0x1ad172ed510>]
```




```
from matplotlib import pyplot
import numpy as np

def fA(x):
    if(x<1):
        return 0
    elif  1 <= x and x < 7:
```

```
        return (x - 1)/6
    elif  7 <= x and x < 10:
        return (10 - x)/3
    elif x >= 10:
        return 0

def fAA(x):
    if(x<1):
        return 1
    elif  1 <= x and x < 7:
        return 1 - ((x - 1)/6)
    elif  7 <= x and x < 10:
        return 1 - ((10 - x)/3)
    elif x >= 10:
        return 1

def fAuAA(x):
    if(x<1):
        return 1
    elif  1 <= x and x < 4:
        return 1 - ((x - 1)/6)
    elif  4 <= x and x < 7:
        return ((x - 1)/6)
    elif  7 <= x and x < 8.5:
        return ((10 - x)/3)
    elif  8.5 <= x and x < 10:
        return 1 - ((10 - x)/3)
    elif x >= 10:
        return 1

x = np.linspace(1, 10, 100)
pyplot.plot(x, [fA(i) for i in x],'--')
pyplot.plot(x, [fAA(i) for i in x],'--')
pyplot.plot(x, [fAuAA(i) for i in x],'-')
```

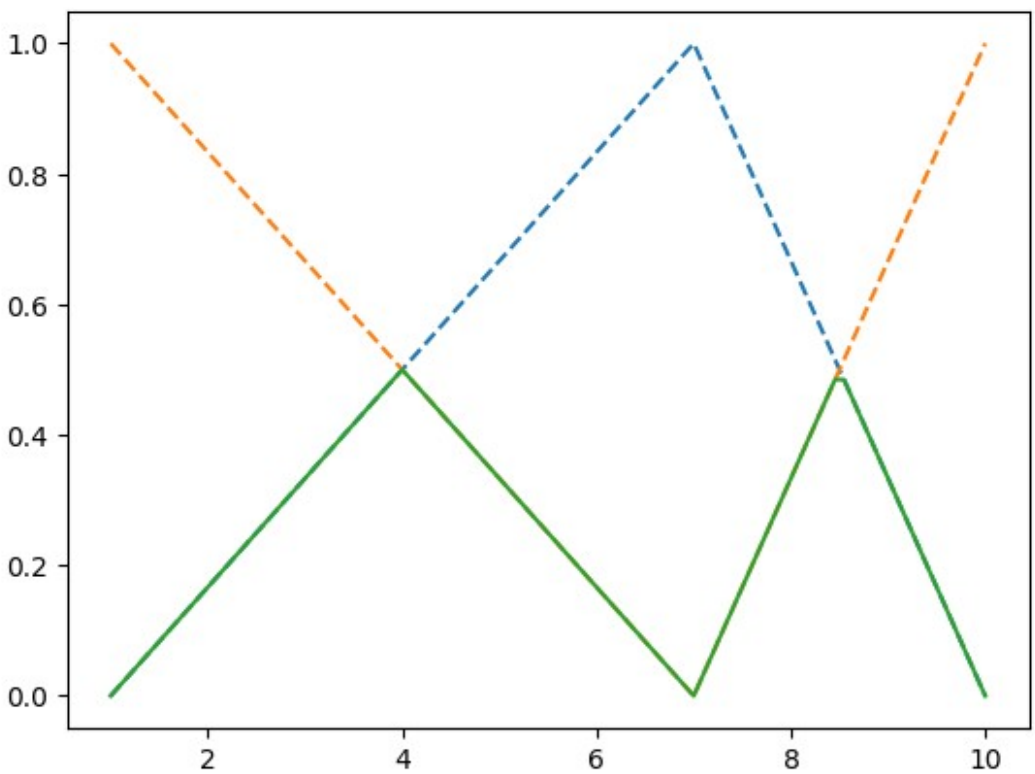
```
[<matplotlib.lines.Line2D at 0x1ad1a7e7310>]
```




```
from matplotlib import pyplot
import numpy as np

def fA(x):
    if(x<1):
        return 0
    elif  1 <= x and x < 7:
        return (x - 1)/6
    elif  7 <= x and x < 10:
        return (10 - x)/3
    elif x >= 10:
        return 0

def fAA(x):
    if(x<1):
        return 1
    elif  1 <= x and x < 7:
        return 1 - ((x - 1)/6)
    elif  7 <= x and x < 10:
        return 1 - ((10 - x)/3)
    elif x >= 10:
        return 1

def fAuAA(x):
    if(x<1):
```

```
        return 0
    elif  1 <= x and x < 4:
        return ((x - 1)/6)
    elif  4 <= x and x < 7:
        return 1 - ((x - 1)/6)
    elif  7 <= x and x < 8.5:
        return 1 - ((10 - x)/3)
    elif  8.5 <= x and x < 10:
        return ((10 - x)/3)
    elif x >= 10:
        return 0

x = np.linspace(1, 10, 100)
pyplot.plot(x, [fA(i) for i in x],'--')
pyplot.plot(x, [fAA(i) for i in x],'--')
pyplot.plot(x, [fAuAA(i) for i in x],'-')
```

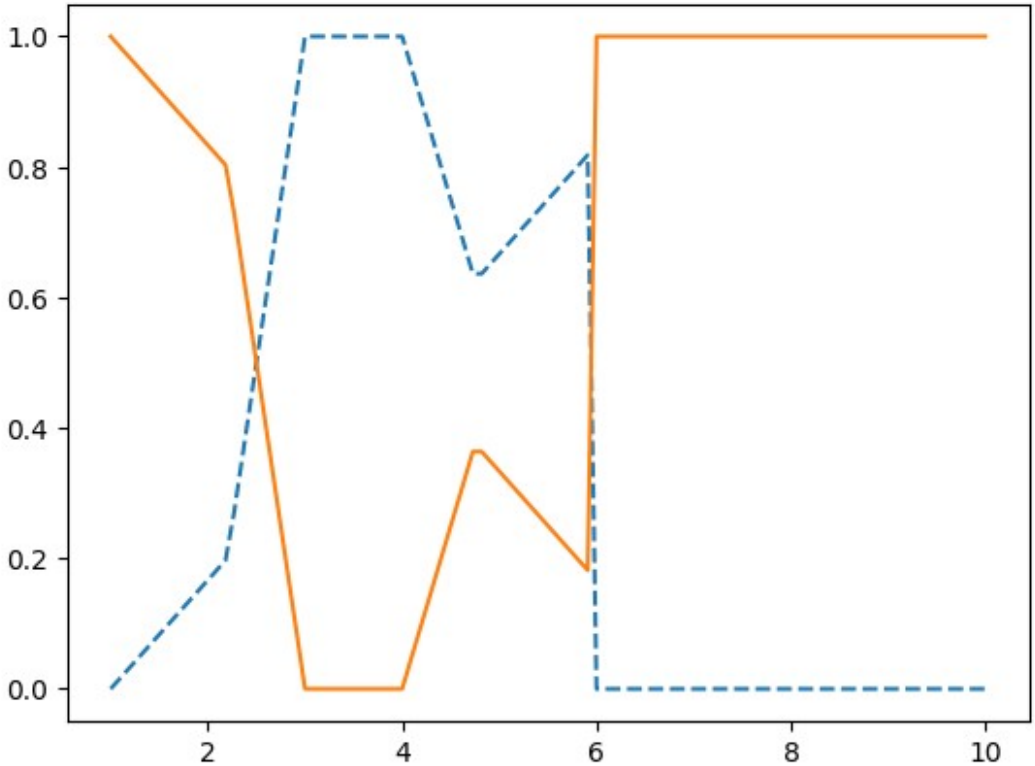
```
[<matplotlib.lines.Line2D at 0x1ad1a852b50>]
```




```
from matplotlib import pyplot
import numpy as np

def fA(x):
    if x < 1:
        return 0
```

```
    elif  1 <= x and x < 2.2:
        return (x - 1)/6
    elif  2.2 <= x and x < 3:
        return (x-2)
    elif  3 <= x and x < 4:
        return 1
    elif  4 <= x and x < 4.75:
        return (6-x)/2
    elif  4.75 <= x and x < 6:
        return (x - 1)/6
    elif x >= 6:
        return 0
def fB(x):
    if x < 1:
        return 1
    elif  1 <= x and x < 2.2:
        return 1-(x - 1)/6
    elif  2.2 <= x and x < 3:
        return 1-(x-2)
    elif  3 <= x and x < 4:
        return 0
    elif  4 <= x and x < 4.75:
        return 1-(6-x)/2
    elif  4.75 <= x and x < 6:
        return 1-(x - 1)/6
    elif x >= 6:
        return 1

x = np.linspace(1, 10, 100)
pyplot.plot(x, [fA(i) for i in x],'--')
pyplot.plot(x, [fB(i) for i in x],'-')
```

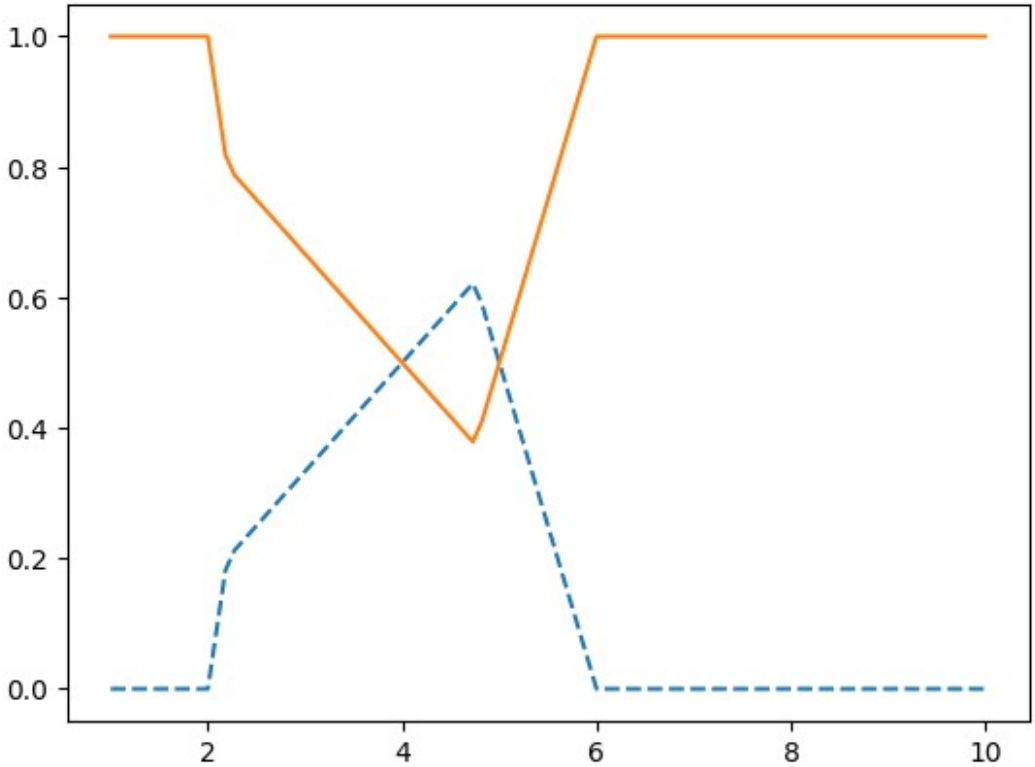
```
[<matplotlib.lines.Line2D at 0x1ad1a88ba50>]
```




```
from matplotlib import pyplot
import numpy as np

def fA(x):
    if x < 2:
        return 0
    elif  2 <= x and x < 2.2:
        return (x -2)
    elif  2.2 <= x and x < 4.75:
        return (x - 1)/6
    elif  4.75 <= x and x < 6:
        return (6-x)/2
    elif x >= 6:
        return 0
def fB(x):
    if x < 2:
        return 1
    elif  2 <= x and x < 2.2:
        return 1-(x -2)
    elif  2.2 <= x and x < 4.75:
        return 1-(x - 1)/6
    elif  4.75 <= x and x < 6:
        return 1-(6-x)/2
    elif x >= 6:
        return 1
```

```
x = np.linspace(1, 10, 100)
pyplot.plot(x, [fA(i) for i in x],'--')
pyplot.plot(x, [fB(i) for i in x],'-')
```

```
[<matplotlib.lines.Line2D at 0x1ad17166ed0>]
```



$supp(A)=¿$

$supp(B)=¿$

$core(A)=7$

$core(A)=[3,4)$

$h(A)=1$

$h(B)=1$

Ejercicio 7. Demuestra las propiedades 1 al 5, 7 y 8 en la Proposición 1.15 de libro de Bede (2013).

Respuestas:

---

1. Associativity

$A \wedge (B \wedge C) = (A \wedge B) \wedge C$

$min(\mu_A, min(\mu_B, \mu_C)) = min(\mu_A, \mu_B, \mu_C) = min(min(\mu_A, \mu_B), \mu_C)$

Caso 1. Si $(\mu_A \leq \mu_B)$ y $(\mu_A \leq \mu_C)$ entonces $\mu_A = \mu_A$

Caso 2. Si $(\mu_A \geq \mu_B)$ y $(\mu_A \geq \mu_C)$ entonces $min(\mu_B, \mu_C) = min(\mu_B, \mu_C)$

Caso 3. Si $(\mu_B \geq \mu_A \geq \mu_C)$ ó $(\mu_C \geq \mu_A \geq \mu_B)$ entonces $\mu_C = \mu_C$ ó $\mu_B = \mu_B$

$A \vee (B \vee C) = (A \vee B) \vee C$

$max(\mu_A, max(\mu_B, \mu_C)) = max(\mu_A, \mu_B, \mu_C) = max(max(\mu_A, \mu_B), \mu_C)$

Caso 1. Si $(\mu_A \leq \mu_B)$ y $(\mu_A \leq \mu_C)$ entonces $max(\mu_B, \mu_C) = max(\mu_B, \mu_C)$

Caso 2. Si $(\mu_A \geq \mu_B)$ y $(\mu_A \geq \mu_C)$ entonces $\mu_A = \mu_A$

Caso 3. Si $(\mu_B \geq \mu_A \geq \mu_C)$ ó $(\mu_C \geq \mu_A \geq \mu_B)$ entonces $\mu_B = \mu_B$ ó $\mu_C = \mu_C$

---

2. Commutativity

$A \wedge B = B \wedge A$

$min(\mu_A, \mu_B) = min(\mu_B, \mu_A)$

Caso 1. Si $(\mu_A \leq \mu_B)$ entonces $\mu_A = \mu_A$

Caso 2. Si $(\mu_A \geq \mu_B)$ entonces $\mu_B = \mu_B$

$A \vee B = B \vee A$

$max(\mu_A, \mu_B) = max(\mu_B, \mu_A)$

Caso 1. Si $(\mu_A \leq \mu_B)$ entonces $\mu_B = \mu_B$

Caso 2. Si $(\mu_A \geq \mu_B)$ entonces $\mu_A = \mu_A$

---

3. Identity

$A \wedge X = A$

$min(\mu_A, 1) = \mu_A$

$A \vee \varnothing = A$

$max(\mu_A, 0) = \mu_A$

---

4. Absorption by ∅ and X

$A \wedge \varnothing = \varnothing$

$min(\mu_A, 0) = 0$

$A \vee X = X$

$max(\mu_A, 1) = 1$

---

5. Idempotence

$A \wedge A = A$

$min(\mu_A, \mu_A) = min(\mu_A) = \mu_A$

$A \vee A = A$

$max(\mu_A, \mu_A) = max(\mu_A) = \mu_A$

---

7. Distributivity

$A \wedge (B \vee C) = (A \wedge B) \vee (A \wedge C)$

$min(\mu_A, max(\mu_B, \mu_C)) = max(min(\mu_A, \mu_B), min(\mu_A, \mu_C))$

Caso 1. Si $(\mu_A \leq \mu_B)$ y $(\mu_A \leq \mu_C)$ entonces $\mu_A = \mu_A$

Caso 2. Si $(\mu_A \geq \mu_B)$ y $(\mu_A \geq \mu_C)$ entonces $max(\mu_B, \mu_C) = max(\mu_B, \mu_C)$

Caso 3. Si $(\mu_B \geq \mu_A \geq \mu_C)$ ó $(\mu_C \geq \mu_A \geq \mu_B)$ entonces $\mu_A = \mu_A$

$A \vee (B \wedge C) = (A \vee B) \wedge (A \vee C)$

$max(\mu_A, min(\mu_B, \mu_C)) = min(max(\mu_A, \mu_B), max(\mu_A, \mu_C))$

Caso 1. Si $(\mu_A \leq \mu_B)$ y $(\mu_A \leq \mu_C)$ entonces $min(\mu_B, \mu_C) = min(\mu_B, \mu_C)$

Caso 2. Si $(\mu_A \geq \mu_B)$ y $(\mu_A \geq \mu_C)$ entonces $\mu_A = \mu_A$

Caso 3. Si $(\mu_B \geq \mu_A \geq \mu_C)$ ó $(\mu_C \geq \mu_A \geq \mu_B)$ entonces $\mu_A = \mu_A$

---

8. Involution

$\overline{\overline{A}}=A$

$\mu_{\overline{\overline{A}}}=1-\mu_{\overline{A}}=1-(1-\mu_A)=1-1+\mu_A=\mu_A$

Ejercicio 8. Demostrar que para cualquier conjunto difuso $A: X \to [0,1)$ se cumple que para todo $x \in X$ (i) $A(x) \cup \overline{A}(x) \geq 0.5$ y (ii) $A(x) \cap \overline{A}(x) \leq 0.5$.

Respuestas:

Sabemos que la suma de las membresías de un conjunto y su complemento es siempre igual a 1.

Por ejemplo, si $A(x)=0.4$ entonces $\overline{A}(x)=1-A(x)=1-0.4=0.6$

(i) $A(x) \cup \overline{A}(x) \geq 0.5$

$A(x) \cup \overline{A}(x) \geq 0.5=max(A(x),\overline{A}(x)) \geq 0.5=max(0.4,0.6) \geq 0.5=0.6 \geq 0.5$

(ii) $A(x) \cap \overline{A}(x) \leq 0.5$

$A(x) \cap \overline{A}(x) \leq 0.5=min(A(x),\overline{A}(x)) \leq 0.5=min(0.4,0.6) \leq 0.5=0.4 \leq 0.5$

Ejercicio 9. Demostrar que para cualquier par de conjuntos difusos $A: X \to [0,1)$ y $B: X \to [0,1)$ se cumple que

1. $(A \cap \overline{B}) \cup (\overline{A} \cap B) \supseteq 0.5 \cap (A \cup B) \cap (\overline{A} \cup \overline{B})$ y
2. $(A \cup \overline{B}) \cap (\overline{A} \cup B) \subseteq 0.5 \cup (A \cap B) \cup (\overline{A} \cap \overline{B})$.

Respuestas:

---

1. $Solución$

$(A \cap \overline{B}) \cup (\overline{A} \cap B) \supseteq 0.5 \cap (A \cup B) \cap (\overline{A} \cup \overline{B})$

Supongamos que $max(A,B)=A$, o sea $A>B$ entonces:

- Caso 1: $A>0.5$ y $B>0.5$
  Implica que $\overline{A}<0.5$ y $\overline{B}<0.5$
  entonces

$max(min(A(x),\overline{B}(x)),min(\overline{A}(x),B(x)))$
$max(\overline{B}(x),\overline{A}(x))=\overline{B}(x)$

$min(0.5,max(A(x),B(x)),max(\overline{A}(x),\overline{B}(x)))\ min(0.5,A(x),\overline{B}(x))=\overline{B}(x)$

$ B(x) \geq B(x)$

- Caso 2: $A<0.5$ y $B<0.5$
  Implica que $\overline{A}>0.5$ y $\overline{B}>0.5$
  entonces

$max(min(A(x),\overline{B}(x)),min(\overline{A}(x),B(x)))$
$max(A(x),B(x))=A(x)$

$min(0.5,max(A(x),B(x)),max(\overline{A}(x),\overline{B}(x)))\ min(0.5,A(x),\overline{B}(x))=A(x)$

$ A(x) \geq A(x)$

- Caso 3: $A>0.5$ y $B<0.5$
  Implica que $\overline{A}<0.5$ y $\overline{B}>0.5$
  entonces

$max(min(A(x),\overline{B}(x)),min(\overline{A}(x),B(x)))=min(A(x),\overline{B}(x))$

$min(0.5,max(A(x),B(x)),max(\overline{A}(x),\overline{B}(x)))$
$min(0.5,A(x),\overline{B}(x))=0.5$

$min(A(x),\overline{B}(x))\geq 0.5$

Por último en caso de que $A=B$

- Si $A(x) > 0.5, B(x) > 0.5 $ Implica que $\overline{A}<0.5$ y $\overline{B}<0.5$ $y\ \overline{A}=\overline{B}$ entonces

$max(min(A(x),\overline{B}(x)),min(\overline{A}(x),B(x)))$
$max(\overline{B}(x),\overline{A}(x))=\overline{A}(x)=\overline{B}(x)$

$min(0.5,max(A(x),B(x)),max(\overline{A}(x),\overline{B}(x)))$
$min(0.5,A(x),\overline{A}(x))=\overline{A}(x)=\overline{B}(x)$

$ \overline A(x) \geq \overline A(x)$

---

1. $Solución$
   $(A \cup \overline{B}) \cap (\overline{A} \cup B) \subseteq 0.5 \cup (A \cap B) \cup (\overline{A} \cap \overline{B})$.

Supongamos que $max(A,B)=A$, o sea $A>B$ entonces:

- Caso 1: $A>0.5$ y $B>0.5$
  Implica que $\overline{A}<0.5$ y $\overline{B}<0.5$
  entonces

$$min\left(max\left(A(x),\overline{B}(x)\right),max\left(\overline{A}(x),B(x)\right)\right)$$
$$min\left(A(x),B(x)\right)=B(x)$$

$$max\left(0.5,min\left(A(x),B(x)\right),min\left(\overline{A}(x),\overline{B}(x)\right)\right)\ max\left(0.5,B(x),\overline{A}(x)\right)=B(x)$$

$ B(x) \leq B(x)$

- Caso 2: $A<0.5$ y $B<0.5$
  Implica que $\overline{A}>0.5$ y $\overline{B}>0.5$
  entonces

$$min\left(max\left(A(x),\overline{B}(x)\right),max\left(\overline{A}(x),B(x)\right)\right)$$
$min(\overline B(x) , \overline A(x)) = \overline A(x)

$$max\left(0.5,min\left(A(x),B(x)\right),min\left(\overline{A}(x),\overline{B}(x)\right)\right)\ max\left(0.5,B(x),\overline{A}(x)\right)=\overline{A}(x)$$

$ \overline A(x) \leq \overline A(x)$

- Caso 3: $A>0.5$ y $B<0.5$
  Implica que $\overline{A}<0.5$ y $\overline{B}>0.5$
  entonces

$$min\left(max\left(A(x),\overline{B}(x)\right),max\left(\overline{A}(x),B(x)\right)\right)=max\left(\overline{A}(x),B(x)\right)$$

$$max\left(0.5,min\left(A(x),B(x)\right),min\left(\overline{A}(x),\overline{B}(x)\right)\right)$$
$$max\left(0.5,B(x),\overline{A}(x)\right)=0.5$$

$$max\left(\overline{A}(x),B(x)\right)\leq 0.5$$

Por último en caso de que $A=B$

- Si $A(x) > 0.5, B(x) > 0.5 $ Implica que $\overline{A}<0.5$ y $\overline{B}<0.5$ $y\ \overline{A}=\overline{B}$ entonces

$$min\left(max\left(A(x),\overline{B}(x)\right),max\left(\overline{A}(x),B(x)\right)\right)$$
$$min\left(A(x),B(x)\right)=A(x)=B(x)$$

$$max\left(0.5,min\left(A(x),B(x)\right),min\left(\overline{A}(x),\overline{B}(x)\right)\right)$$
$$max\left(0.5,A(x),\overline{A}(x)\right)=A(x)=B(x)$$

$ A(x) \leq A(x)$

Ejercicio 10. Denotamos por T_L y S_L la t-norma y t-conomar de Łukasiewicz y están definidas como

$$x T_L y = (x + y - 1) \vee 0$$

y

$$x S_L y = (x + y) \wedge 1.$$

Demostrar que T_L y S_L es una t-norma y t-conorma, respectivamente.

Respuestas:

---

1. Demostrar que T_L es una t-norma

(i) Boundary condition:

$x T_L y = (x + y - 1) \vee 0$

$x T_L 1 = (x + 1 - 1) \vee 0 = x \vee 0 = x$

$0 T_L 0 = (0 + 0 - 1) \vee 0 = -1 \vee 0 = 0$

(ii) Monotonicity:

Si $y \leq z$ entonces $x T_L y \leq x T_L z$

$x T_L y = (x + y - 1) \vee 0$

$x T_L z = (x + z - 1) \vee 0$

$(x + y - 1) \vee 0 \leq (x + z - 1) \vee 0$

$(x + y - 1) \leq (x + z - 1)$

(iii) Commutativity:

$x T_L y = (x + y - 1) \vee 0 = (y + x - 1) \vee 0 = y T_L x$

(iv) Associativity:

$x T_L (y T_L z) = (x + y T_L z - 1) \vee 0 = (x + ((y + z - 1) \vee 0) - 1) \vee 0 = (((x + y - 1) \vee 0) + z - 1) \vee 0 = (x T_L y + z - 1) \vee 0 = ($

---

2. Demostrar que S_L es una t-conorma

(i) Boundary condition:

$x S_L y = (x + y) \wedge 1$

$x S_L 0 = (x + 0) \wedge 1 = x \wedge 1 = x$

$x S_L 1 = (x + 1) \wedge 1 = 1$

(ii) Monotonicity:

Si $y \leq z$ entonces $x\, S_L\, y \leq x\, S_L\, z$

$x\, S_L\, y = (x+y) \wedge 1$

$x\, S_L\, z = (x+z) \wedge 1$

$(x+y) \wedge 1 \leq (x+z) \wedge 1$

$(x+y) \leq (x+z)$

(iii) Commutativity:

$x\, S_L\, y = (x+y) \wedge 1 = (y+x) \wedge 1 = y\, S_L\, x$

(iv) Associativity:

$x\, S_L (y\, S_L\, z) = (x + y\, S_L\, z) \wedge 1 = (x + ((y+z) \wedge 1)) \wedge 1 = (((x+y) \wedge 1) + z) \wedge 1 = (x\, S_L\, y + z) \wedge 1 = (x\, S_L\, y) S_L\, z$

Ejercicio 11. Considera los conjuntos difusos $A, B : R^{+¿ \to [0,1]¿}$ definidos como $A(x) = 1/(1+x^2)$ y $B(x) = 1/10^x$. Escribe formalmente como funciones $A\, T_L\, B$, $A\, S_L\, B$, $A\, T_G\, B$, $A\, S_G\, B$, donde T_L y S_L son la t-norma y t-conorma de Łukasiewicz y T_G y S_G son la t-norma y t-conorma de Goguen. Presenta una gráfica de cada uno.

Respuestas:

(i) $x\, T_L\, y = (x+y-1) \vee 0$

$A\, T_L\, B(x) = T_L(A(x), B(x)) = ((1/(1+x^2)) + (1/10^x) - 1) \vee 0$

(ii) $x\, S_L\, y = (x+y) \wedge 1$

$A\, S_L B(x) = S_L(A(x), B(x)) = ((1/(1+x^2)) + (1/10^x)) \wedge 1$

(iii) $x\, T_G\, y = x \cdot y$

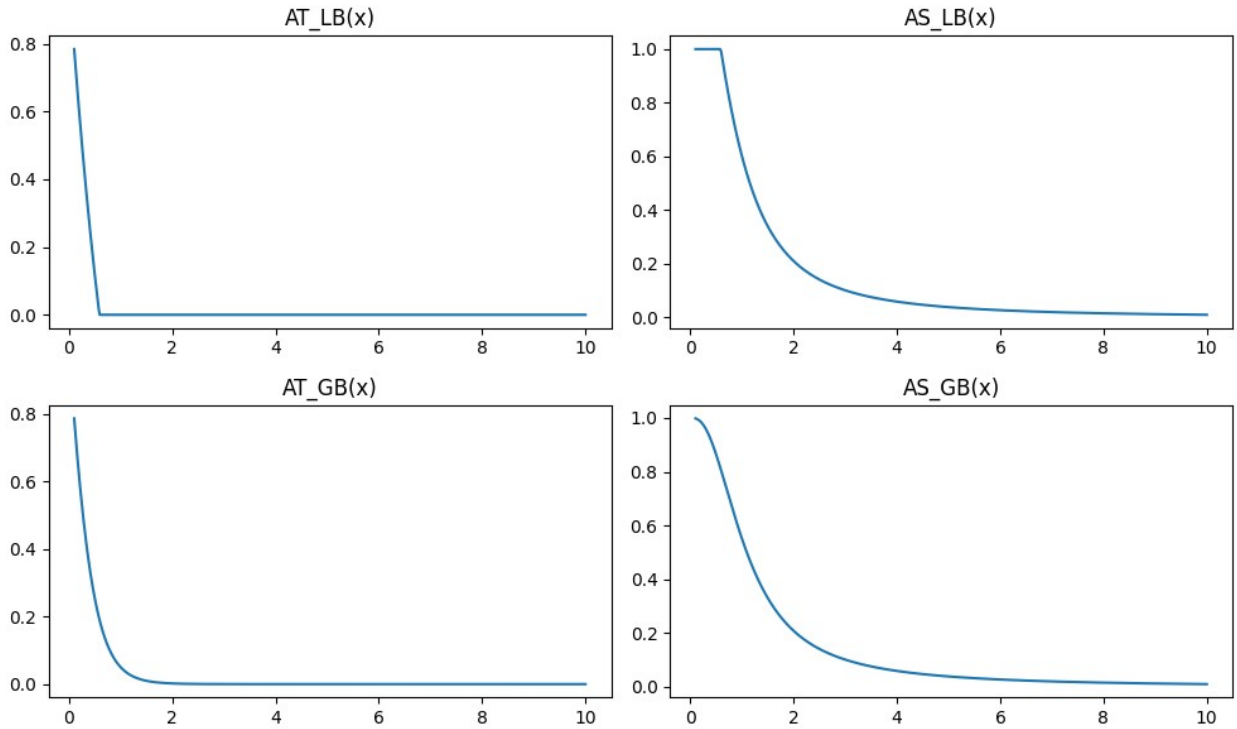
$A\, T_G B(x) = T_G(A(x), B(x)) = (1/(1+x^2)) \cdot (1/10^x)$

(iv) $x\, S_G\, y = x + y - x\, y$

$A\, S_G B(x) = S_G(A(x), B(x)) = (1/(1+x^2)) + (1/10^x) - (1/(1+x^2))(1/10^x)$

```
import matplotlib.pyplot as plt

import numpy as np
```

```python
def A(x):
    return 1 / (1 + x**2)

def B(x):
    return 1 / 10**x

def AT_LB(x):
    return np.maximum((A(x) + B(x) - 1), 0)

def AS_LB(x):
    return np.minimum((A(x) + B(x)), 1)

def AT_GB(x):
    return A(x) * B(x)

def AS_GB(x):
    return A(x) + B(x) - (A(x) * B(x))

x_values = np.linspace(0.1, 10, 500)  # Por ejemplo, de 0.1 a 10 con
500 puntos

AT_LB_values = AT_LB(x_values)
AS_LB_values = AS_LB(x_values)
AT_GB_values = AT_GB(x_values)
AS_GB_values = AS_GB(x_values)

plt.figure(figsize=(10, 6))

plt.subplot(2, 2, 1)
plt.plot(x_values, AT_LB_values)
plt.title('AT_LB(x)')

plt.subplot(2, 2, 2)
plt.plot(x_values, AS_LB_values)
plt.title('AS_LB(x)')

plt.subplot(2, 2, 3)
plt.plot(x_values, AT_GB_values)
plt.title('AT_GB(x)')

plt.subplot(2, 2, 4)
plt.plot(x_values, AS_GB_values)
plt.title('AS_GB(x)')

plt.tight_layout()
plt.show()
```


AT_LB(x)
AS_LB(x)
AT_GB(x)
AS_GB(x)
0.0
0.2
0.4
0.6
0.8
1.0
0
2
4
6
8
10